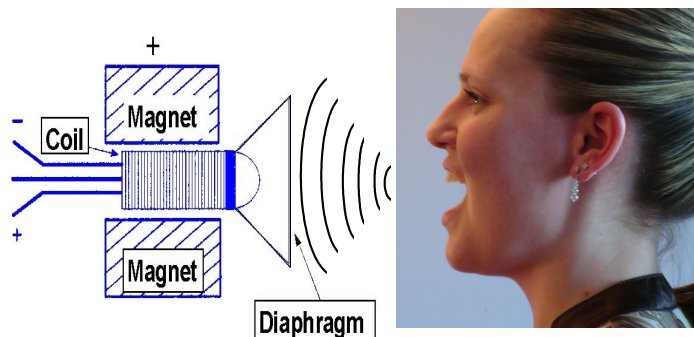


Proper Microphone Techniques For Vocalist.

One of the greatest problems in performing that vocalist have is the lack of understanding on using microphones. The first thing you will have to get use to that the your voice dose not sound the same way you hear it as it dose to someone listening to you. What dose this have to do with microphones you say? Everything. The greatest shock to most new singers is when they step up to a mic and hear something totally different then what they hear when they don't use a mic. To most people it is a bit unnerving and they tend to shy away from the microphone thinking that it is not doing a very good job of reproducing your voice. Well the fact is, it is, and the way you hear your voice through the monitors is how other people hear you, and you need to embrace that fact.

What happens is your head acts like a speaker cabinet, and your mouth, ears, and nose act like ports in the cabinet. When you sing your body vibrates producing frequencies to help your vocal cords produce the overall sound that is projected from your mouth. Your ears pick up some of the vibrations from your body adding a tonal quality to your voice that only you can hear. This has a tendency to mess with your mind if you are not accustomed to hearing your voice from an external source. The best way to get over this is to have some one record you singing, listen to it for this is what everyone else hears when you sing. Now embrace it. Woman have a tendency to except their voice as others hear it more then men, due to the fact that mens voices carry more bass frequencies. The sooner you get over trying to make everyone hear what you hear the more confident you will feel on stage. When you are confident in how your voice sounds you are no longer afraid to use the microphone. Confident singers are able to project their feelings through their music, and it is then that a microphone can be your best friend not your enemy.

So now you have come to love your voice as others do, it is time to learn how to allow your body, voice and microphone become one instrument. First a crash course on performance vocal microphones. As you sing you produce sound waves that move through the air and hit the microphone's diaphragm and cause it to move back and forth creating an electrical current that is sent to the sound board.



If you sing very loudly and really close to the mic you could cause the diaphragm to bottom out and cause distortion with in the sound, but if you sing to softly and the mic is to far away then you may not get the diaphragm to move properly. A proper way to hold your mic should be between two and three fingers between your mouth and the mic.



As you sing quietly you move the mic closer and as you project your voice you move it away slightly



I have met some singers who where taught to sing with the mic against their chin. This is *not* how you should use a mic for two reasons. One is you when sing with the mic against your chin it rubs against your skin causing unwanted noise, and two you are projecting over the mic and not into it.



A microphone has a pickup pattern that causes frequencies and volume to be lost when it is off axis. On axis is when the sound being picked up by the mic is being

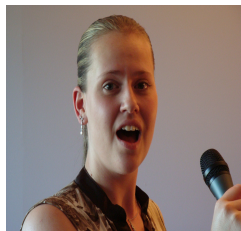
projected directly into the mic. This gives the mic the best response to all the frequencies that is being produced. When it is off axis the higher frequencies are attenuated (turned down) giving an unmusical muddy sound.



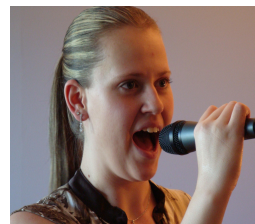
0° is considered on axis and as you go off axis you can see how the signal attenuates. Higher frequencies attenuate at a faster rate.

The only reason a person might be taught to rest the mic against their chin is because the sound person did not want to invest the time to train them to use it properly or they didn't know any better. Many new singers have a problem keeping the mic on axis with their mouth, but this is not the way to deal with it. If you are new to singing on stage use a stand until you get use to it.

Wrong



Right



When it comes time for sound check make sure you put your all into the sound check, and don't lay back in projecting your voice. This is the time the sound person is setting the gain control for the mic, and you want it gained properly. If you just project your voice at $\frac{3}{4}$ of your normal volume, when it comes time for the real thing you will overload the board and cause the signal to distort. So give them the real thing at sound check.

Take the time to learn to use a mic well and it will make the difference between a good performance and an excellent one. And one last thing, use a mic that works well for you. Not all mic are created equal, so you can not expect them sound the same. Some mics have a more pronounced proximity effect (adds a boost to the bass frequencies) as you bring the microphone gets closer to your mouth. This can make a thin voice sound more fuller, but can make a solid voice muddy. Ask around and find out what people who have a similar voice are using, and try it out for yourself. If you are

serious about singing, buy your own microphone that fits your voice and singing style. Most music stores will let you try them out in the store, so don't be afraid to ask to try different ones. A good microphone will set you back \$150.00 to \$250.00, but it will be an investment you will not regret.

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